

Lab 10: Premium Fabric Capacity

Lab 10: Premium, Fabric, and Capacity-Aware Architecture

Lab summary

These labs and demos help learners compare capacity options and evaluate advanced architecture features. Most hands-on activities are optional because they depend on licensing, capacity, cloud, tenant settings, and customer policy.

Novice-friendly how-to guide

Compare capacity options

1. Start with the workload requirements: user count, model size, refresh frequency, sharing needs, and governance needs.
2. List which requirements work with Pro, PPU, Premium, or Fabric capacity.
3. Mark features that require tenant, license, capacity, or cloud validation.
4. Choose a Gov-safe fallback when a feature is not validated.
5. Record the recommendation and the reason for it.

Review capacity metrics conceptually

1. Identify the capacity or workspace that would be monitored.
2. Review which metrics matter: refresh duration, query duration, throttling, memory, and user activity.
3. If the capacity metrics app is unavailable, document what evidence the admin team would need to provide.

Azure Government readiness

Fabric capacity, Direct Lake, OneLake, Lakehouse, Warehouse, Semantic Link, and autoscale are **Commercial-focused / Verify for Gov**. XMLA, paginated reports, large semantic model settings, and capacity metrics are **Verify for Gov**.

Prerequisites

- Power BI Service access.
- Optional Premium/PPU/Fabric workspace.
- Optional XMLA-compatible tooling.
- Optional Power BI Report Builder.
- Optional capacity metrics app access.
- Optional Fabric workspace with validated feature availability.

Exercise 1: Licensing and capacity comparison

Objective: Choose architecture options based on workload requirements.

Tasks

1. Review the licensing and capacity comparison table.
2. Define a workload scenario.
3. Identify user count, model size, refresh, latency, and governance needs.
4. Select possible capacity options.
5. Document required validation for Azure Government.

Expected result

Learners can recommend an option with caveats and validation needs.

Exercise 2: XMLA endpoint connection where available

Azure Government note: XMLA endpoint is **Verify for Gov**. Validate capacity, tenant settings, workspace configuration, tooling, and cloud support before making this hands-on.

Tasks when available

1. Confirm XMLA endpoint setting.
2. Connect with an approved tool.
3. Inspect model metadata.
4. Document allowed read/write operations.

Alternate path

Review XMLA use cases conceptually and document validation requirements.

Exercise 3: Paginated report where available

Azure Government note: Paginated reports are **Verify for Gov**. Validate licensing, workspace support, Report Builder use, and cloud availability.

Tasks when available

1. Open or create a simple paginated report.
2. Connect to approved sample data.
3. Add a table and parameter.
4. Publish where supported.
5. Review export behavior.

Alternate path

Discuss scenarios where paginated reports are preferred over interactive reports.

Exercise 4: Large semantic model settings where available

Azure Government note: Large semantic models are **Verify for Gov**. Validate capacity and tenant settings.

Tasks when available

1. Review workspace capacity.
2. Review semantic model settings.
3. Identify model size constraints.
4. Document refresh and memory considerations.

Optional commercial lab: Direct Lake

Azure Government note: Direct Lake is **Commercial-focused / Verify for Gov.** Treat as optional unless validated.

Tasks when available

1. Review source data in OneLake/Lakehouse/Warehouse.
2. Create or inspect a Direct Lake semantic model.
3. Compare with Import and DirectQuery concepts.
4. Document limitations and fallback path.

Optional commercial lab: OneLake/Lakehouse/Warehouse integration

Azure Government note: OneLake, Lakehouse, and Warehouse are **Commercial-focused / Verify for Gov.**

Tasks when available

1. Review workspace items.
2. Identify how data is stored and exposed to Power BI.
3. Compare Lakehouse and Warehouse roles.
4. Document architecture and governance considerations.

Optional commercial lab: Semantic Link

Azure Government note: Semantic Link is **Commercial-focused / Verify for Gov.**

Tasks when available

1. Review Semantic Link use cases.
2. Connect to semantic model metadata from a notebook where supported.
3. Document governance and security considerations.

Exercise 5: Capacity metrics and throttling concepts

Azure Government note: Capacity metrics are **Verify for Gov.** Validate app availability, capacity type, permissions, and telemetry.

Tasks when available

1. Open capacity metrics app.
2. Review interactive and background workload indicators.
3. Identify refresh pressure.
4. Identify throttling symptoms.
5. Document operational follow-up.

Alternate path

Review capacity metrics conceptually and map likely symptoms to operational actions.

Validation checklist

- ☐ Capacity comparison completed.
- ☐ XMLA endpoint marked **Verify for Gov.**
- ☐ Paginated reports marked **Verify for Gov.**
- ☐ Large semantic models marked **Verify for Gov.**
- ☐ Direct Lake marked **Commercial-focused / Verify for Gov.**
- ☐ OneLake/Lakehouse/Warehouse marked **Commercial-focused / Verify for Gov.**
- ☐ Semantic Link marked **Commercial-focused / Verify for Gov.**
- ☐ Autoscale marked **Commercial-focused / Verify for Gov.**
- ☐ Capacity metrics marked **Verify for Gov.**
- ☐ Gov-safe alternate architecture documented.